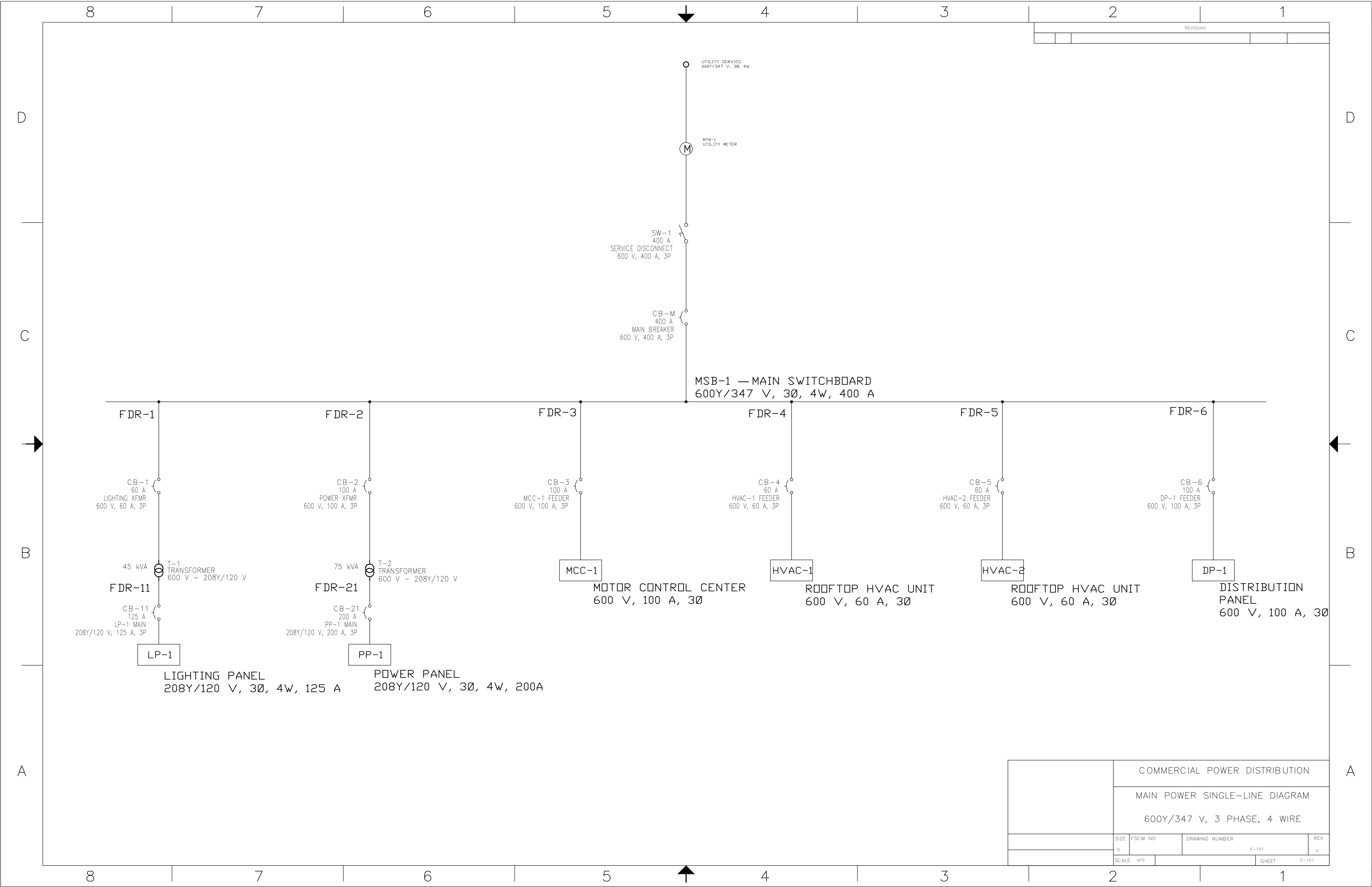
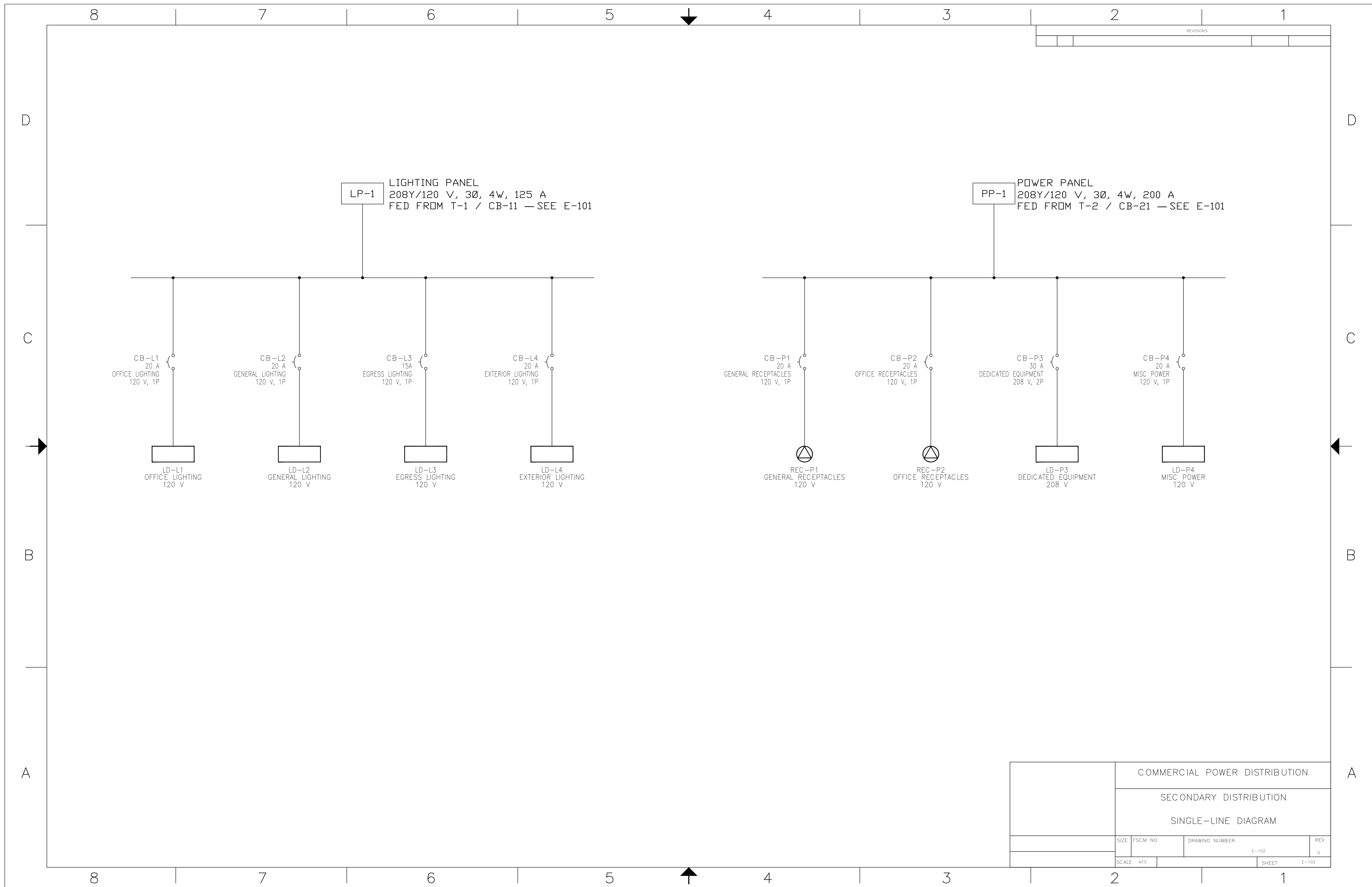
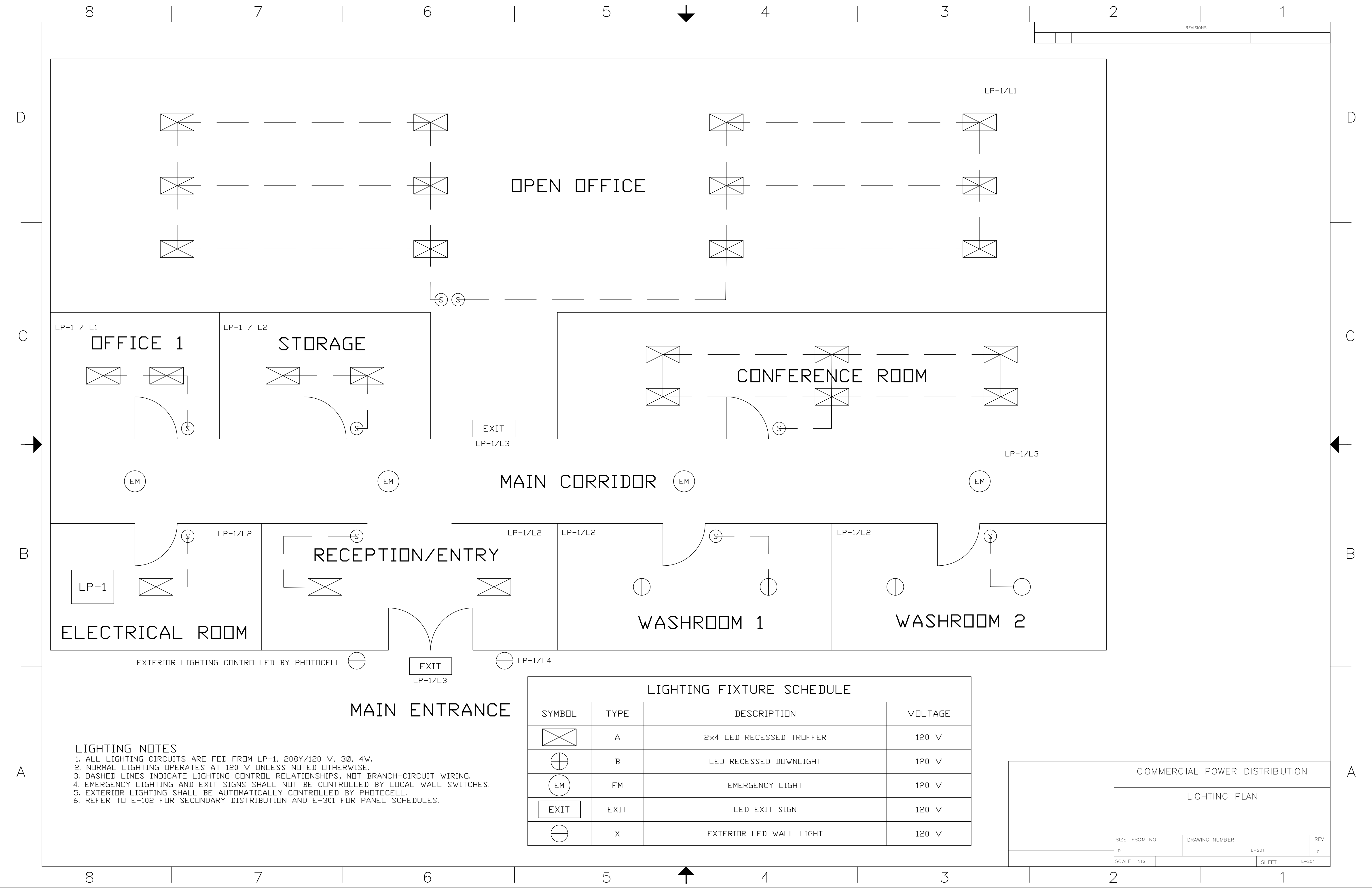




REVISIONS			

	COMMERCIAL POWER DISTRIBUTION			
	MAIN POWER SINGLE-LINE DIAGRAM			
	600Y/347 V, 3 PHASE, 4 WIRE			
	SIZE	FSCM NO	DRAWING NUMBER	REV
	D		E-101	0
	SCALE	NTS	SHEET	E-101





REVISIONS			

LP-1/L1

OPEN OFFICE

LP-1 / L1

OFFICE 1

LP-1 / L2

STORAGE

CONFERENCE ROOM

EXIT
LP-1/L3

LP-1/L3

MAIN CORRIDOR

LP-1/L2

RECEPTION/ENTRY

LP-1/L2

LP-1/L2

LP-1/L2

WASHROOM 1

WASHROOM 2

ELECTRICAL ROOM

EXTERIOR LIGHTING CONTROLLED BY PHOTOCELL

EXIT
LP-1/L3

LP-1/L4

MAIN ENTRANCE

LIGHTING FIXTURE SCHEDULE

SYMBOL	TYPE	DESCRIPTION	VOLTAGE
	A	2x4 LED RECESSED TROFFER	120 V
	B	LED RECESSED DOWNLIGHT	120 V
	EM	EMERGENCY LIGHT	120 V
	EXIT	LED EXIT SIGN	120 V
	X	EXTERIOR LED WALL LIGHT	120 V

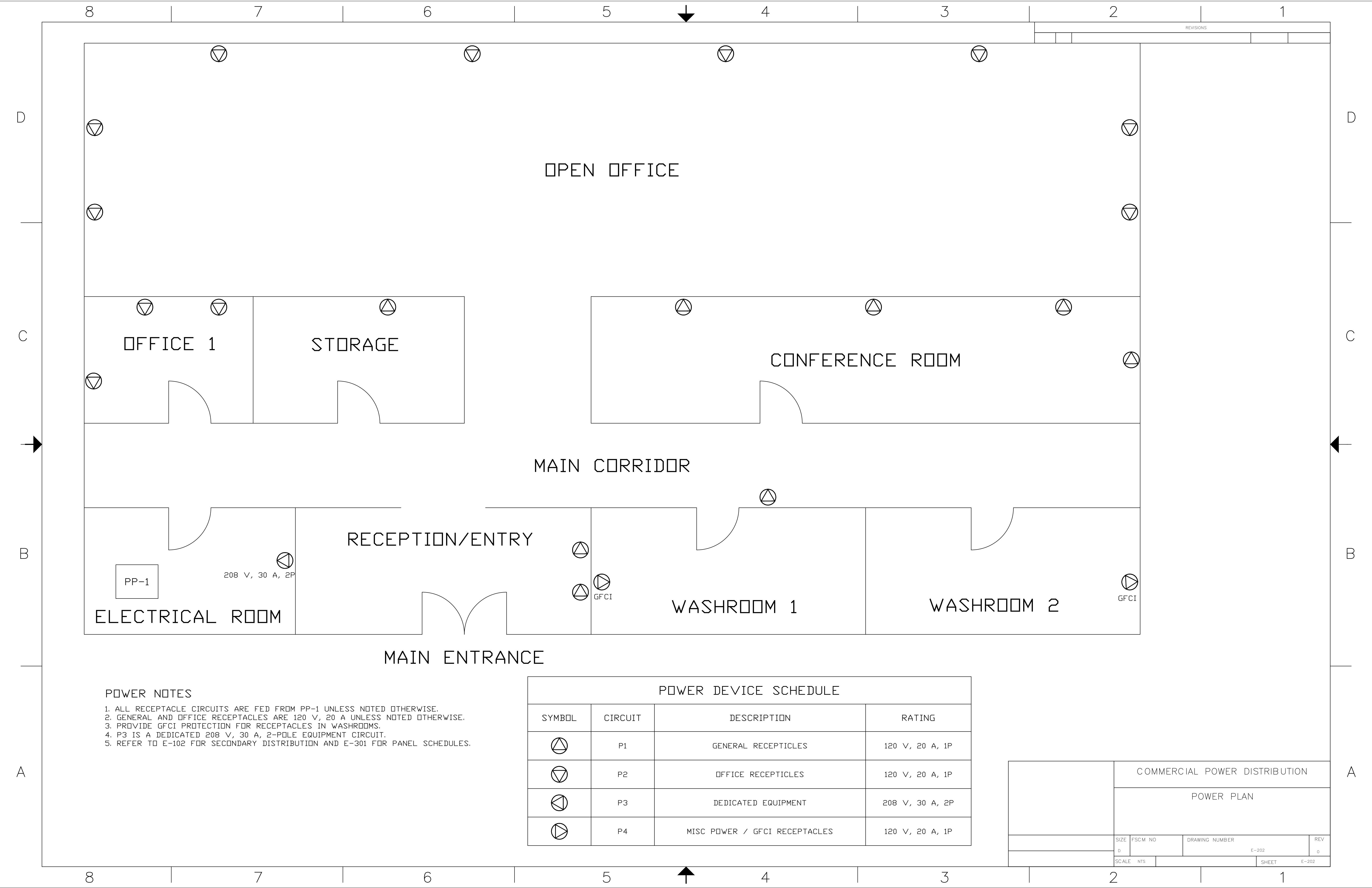
COMMERCIAL POWER DISTRIBUTION

LIGHTING PLAN

LIGHTING NOTES

1. ALL LIGHTING CIRCUITS ARE FED FROM LP-1, 208Y/120 V, 3Ø, 4W.
2. NORMAL LIGHTING OPERATES AT 120 V UNLESS NOTED OTHERWISE.
3. DASHED LINES INDICATE LIGHTING CONTROL RELATIONSHIPS, NOT BRANCH-CIRCUIT WIRING.
4. EMERGENCY LIGHTING AND EXIT SIGNS SHALL NOT BE CONTROLLED BY LOCAL WALL SWITCHES.
5. EXTERIOR LIGHTING SHALL BE AUTOMATICALLY CONTROLLED BY PHOTOCELL.
6. REFER TO E-102 FOR SECONDARY DISTRIBUTION AND E-301 FOR PANEL SCHEDULES.

SIZE	FSCM NO	DRAWING NUMBER	REV
D		E-201	0
SCALE	NTS	SHEET	E-201



POWER NOTES

- 1. ALL RECEPTACLE CIRCUITS ARE FED FROM PP-1 UNLESS NOTED OTHERWISE.
- 2. GENERAL AND OFFICE RECEPTACLES ARE 120 V, 20 A UNLESS NOTED OTHERWISE.
- 3. PROVIDE GFCI PROTECTION FOR RECEPTACLES IN WASHROOMS.
- 4. P3 IS A DEDICATED 208 V, 30 A, 2-POLE EQUIPMENT CIRCUIT.
- 5. REFER TO E-102 FOR SECONDARY DISTRIBUTION AND E-301 FOR PANEL SCHEDULES.

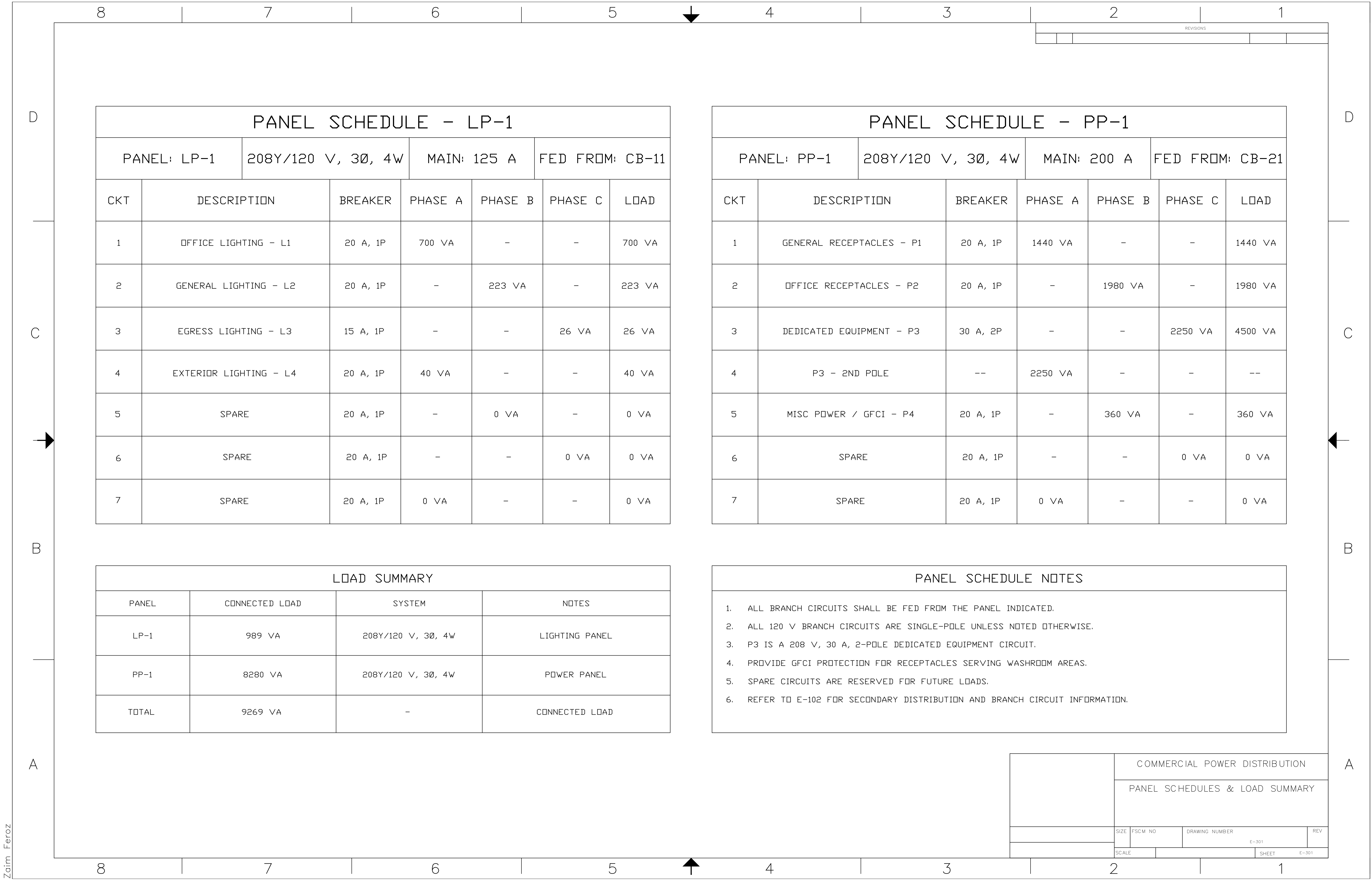
POWER DEVICE SCHEDULE

SYMBOL	CIRCUIT	DESCRIPTION	RATING
	P1	GENERAL RECEPTACLES	120 V, 20 A, 1P
	P2	OFFICE RECEPTACLES	120 V, 20 A, 1P
	P3	DEDICATED EQUIPMENT	208 V, 30 A, 2P
	P4	MISC POWER / GFCI RECEPTACLES	120 V, 20 A, 1P

COMMERCIAL POWER DISTRIBUTION

POWER PLAN

		SIZE	FSCM NO	DRAWING NUMBER	REV
		D		E-202	0
		SCALE	NTS	SHEET	E-202



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REVISIONS

PANEL SCHEDULE - DP-1

PANEL: DP-1		600 V, 3Ø, 4W		MAIN: 100 A		FED FROM: CB-6	
CKT	DESCRIPTION	BREAKER	PHASE A	PHASE B	PHASE C	LOAD	
1	EWH-1 - D1	30A, 3P	6000 VA	-	-	18000 VA	
2	D1 - 2ND POLE	--	-	6000 VA	-	--	
3	D1 - 3RD POLE	--	-	-	6000 VA	--	
4	UH-1 - D2	20 A, 3P	4000 VA	-	-	12000 VA	
5	D2 - 2ND POLE	--	-	4000 VA	-	--	
6	D2 - 3RD POLE	--	-	-	4000 VA	--	
7	UH-2 - D3	20 A, 3P	4000 VA	-	-	12000 VA	
8	D3 - 2ND POLE	--	-	4000 VA	-	--	
9	D3 - 3RD POLE	--	-	-	4000 VA	--	
10	SPARE - D4	30A, 3P	0 VA	-	-	0 VA	
11	D4 - 2ND POLE	--	-	0 VA	-	--	
12	D4 - 3RD POLE	30A, 3P	-	-	0 VA	--	

DP-1 LOAD CALCULATION

LOAD	CONNECTED LOAD	VOLTAGE	CURRENT
EWH-1	18 kVA	600 V, 3Ø	17.3 A
UH-1	12 kVA	600 V, 3Ø	11.5 A
UH-2	12 kVA	600 V, 3Ø	11.5 A

DP-1 BRANCH SINGLE-LINE

DP-1 CALCULATION SUMMARY & NOTES

TOTAL CONNECTED LOAD = 18 + 12 + 12 = 42 kVA
CURRENT = LOAD / (SQRT(3) × VOLTAGE)
I = 42,000 / (SQRT(3) × 600)
I = 40.4 A
DP-1 MAIN RATING = 100 A

1. DP-1 IS FED FROM MSB-1 THROUGH CB-6.
2. ALL ACTIVE DP-1 BRANCH LOADS ARE 600 V, 3-PHASE.
3. D1 SERVES EWH-1, 18 kW.
4. D2 AND D3 SERVE UH-1 AND UH-2, 12 kW EACH.
5. D4 IS RESERVED AS A 30 A, 3-POLE SPARE.
6. REFER TO E-101 FOR MAIN DISTRIBUTION.
7. EWH = ELECTRIC WATER HEATER.
8. UH = UNIT HEATER.

COMMERCIAL POWER DISTRIBUTION

DP-1 PANEL SCHEDULE & LOAD DETAIL

SIZE	FSCM NO	DRAWING NUMBER	REV
		E-302	0
SCALE	NTS	SHEET	E-302

8

7

6

5

4

3

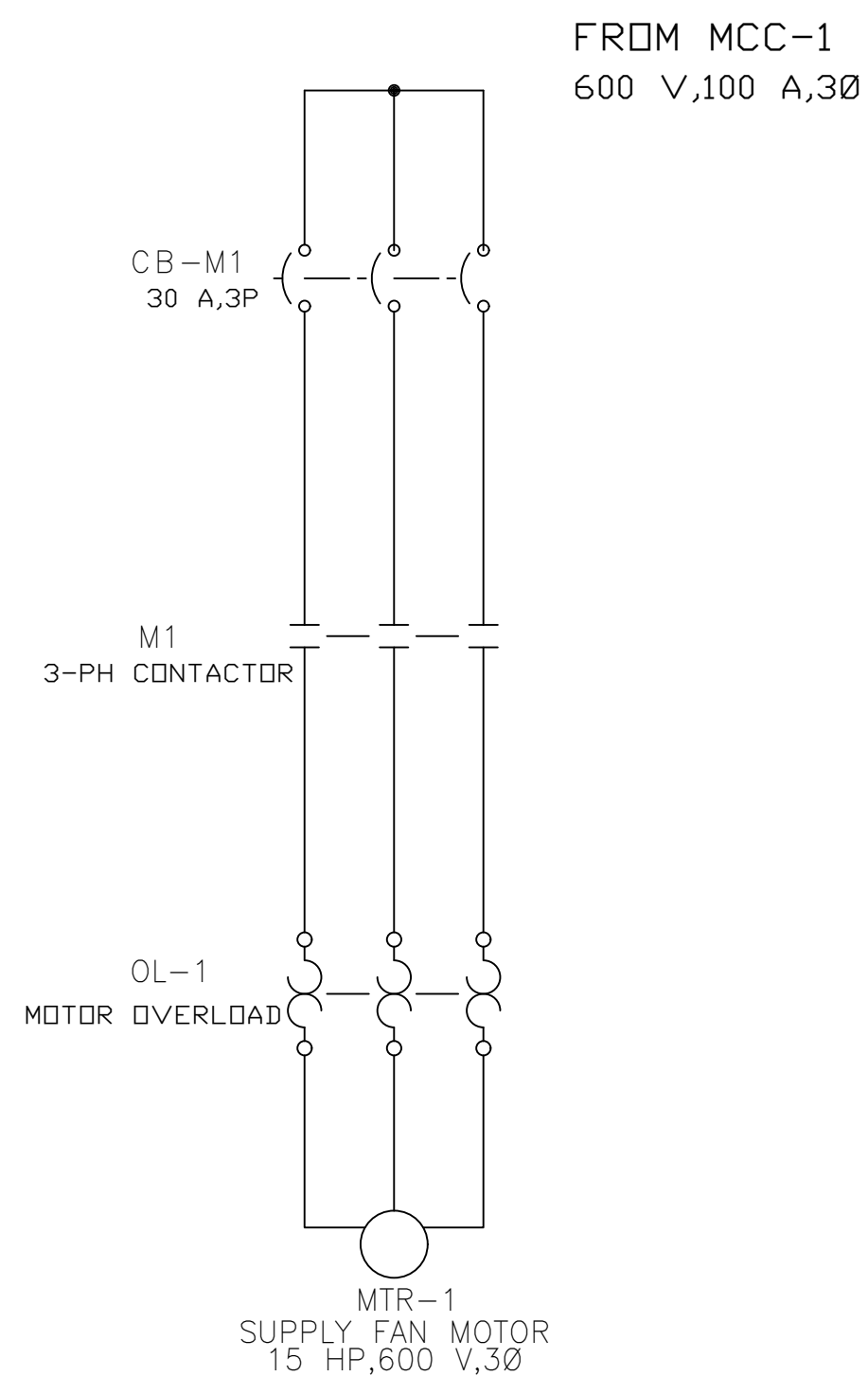
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1

REVISIONS

D

MCC-1 MOTOR POWER CIRCUIT



C

B

MCC-1 EQUIPMENT SCHEDULE

TAG	EQUIPMENT	RATING / DESCRIPTION
CB-M1	CIRCUIT BREAKER	30 A,3P,600 V
M1	MOTOR STARTER	3-PH CONTACTOR
OL-1	OVERLOAD RELAY	MOTOR OVERLOAD PROTECTION
MTR-1	SUPPLY FAN MOTOR	15 HP,600 V,3PH
CPT-1	CONTROL TRANSFORMER	600/120 V
SS-1	HQA SELECTOR SWITCH	HAND-OFF-AUTO
PB-STOP	STOP PUSH BUTTON	NC CONTACT
PB-START	START PUSH BUTTON	ND CONTACT
BAS-1	BAS RUN CONTACT	ND CONTACT
PL-RUN	GREEN PILOT LIGHT	RUN INDICATION
PL-FAULT	RED PILOT LIGHT	FAULT INDICATION

A

8

7

6

5

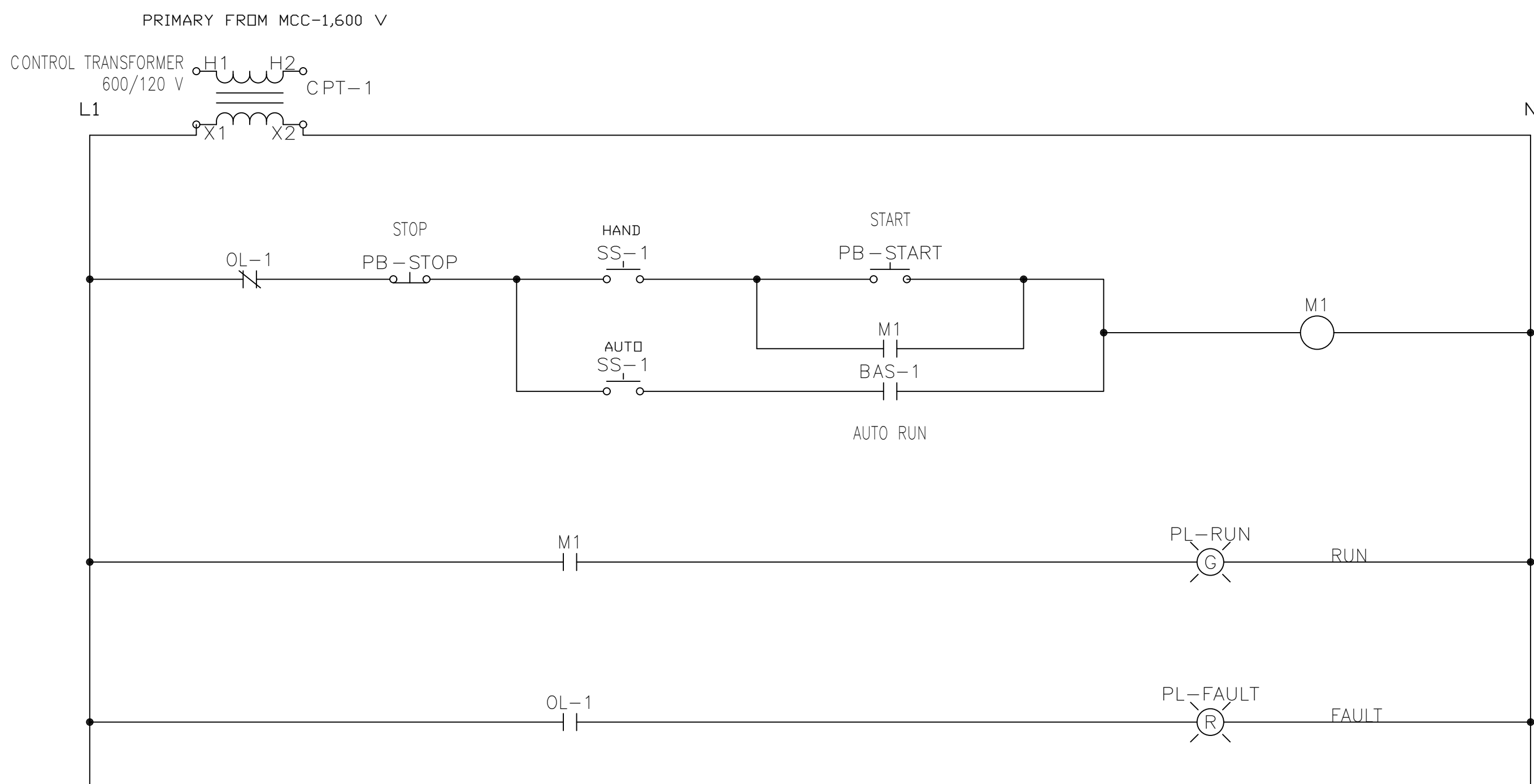
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3

2

1

120 V MOTOR CONTROL CIRCUIT



D

C

B

A

SEQUENCE OF OPERATION & NOTES

- CONTROL POWER SHALL BE SUPPLIED FROM CPT-1,600/120 V.
- OL-1 AND PB-STOP SHALL INTERRUPT M1 IN BOTH HAND AND AUTO MODES.
- IN HAND MODE,PB-START ENERGIZES M1; M1 AUX CONTACT PROVIDES SEAL-IN.
- IN AUTO MODE,BAS-1 RUN COMMAND SHALL ENERGIZE M1.
- IN OFF POSITION,SS-1 SHALL PREVENT M1 FROM ENERGIZING.
- WHEN M1 ENERGIZES,THE STARTER CONTACTS CLOSE AND MTR-1 OPERATES.
- PL-RUN SHALL ILLUMINATE WHEN M1 IS ENERGIZED.
- AN OL-1 TRIP SHALL DE-ENERGIZE M1 AND ILLUMINATE PL-FAULT.
- MTR-1: SUPPLY FAN MOTOR,15 HP,600 V,3-PHASE.

COMMERCIAL POWER DISTRIBUTION

MCC-1 MOTOR POWER

& CONTROL SCHEMATIC

SIZE	FSCM NO	DRAWING NUMBER	REV
		E-401	0

SCALE	NTS	SHEET	E-401
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